

Pyramid Problems · Variables — English Worksheet

Topic: Variables · **Course:** Pyramid Problems · **Time:** about 45 minutes

1 · Predict 🧠

Read each set of steps. Write what you think will happen.

```
set size to 5
repeat size times:
  place block down
  move forward
```

How many blocks does the agent place?


```
set size to 3
repeat size times:
  place block down
  move forward
set size to 6
repeat size times:
  place block down
  move forward
```

How long is the full line of blocks? Why?


```
set height to 4
repeat height times:
  place block down
  move up by 1
```

What does the agent build? How tall?

2 · Spot the Bug 🐛

Each code block below is broken. Rewrite it so it works, then explain the fix.

Bug A — The agent is supposed to build a row of 7 blocks using a variable called `size`.

```
repeat 7 times:  
  place block down  
  move forward
```

Write the fixed code using a variable:

Why is using a variable better here?

Bug B — The agent is supposed to build a row of `size` blocks. The size should be 5.

```
set size to 5  
repeat 3 times:  
  place block down  
  move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — The agent is supposed to build a tower that is `height` blocks tall, where `height` is 6.

```
set height to 6
repeat height times:
  place block down
  move forward
```

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Show Your Work 📷 🎥

Switch to your homework world. Build a **size-5 wall** with a **size** variable, then change it to **12** and compare.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Name the parts.

Fill the blanks:

Today I built _____.

I built it using this code: _____.

In this code I used _____.

The hardest part was _____.

That part was hard because _____.

The most fun part was _____.

Something new I learned was _____.

Write your lines here, then say them in your video.

Submit 

Send this worksheet + your video to teacher on KakaoTalk.